Cover Letter — Micron Technology

Dear Hiring Manager, I am excited to apply for the Design and Verification Engineer, Pathfinding position at Micron Technology. As a Physics and Mathematics student at The College of Idaho with strong experience in quantitative modeling, machine learning, and analytical problem solving, I am eager to contribute to Micron's work in next-generation memory technologies and advanced verification systems. My academic background has provided me with a strong foundation in mathematical modeling, physics, computational analysis, and systems-level thinking. Through both coursework and independent technical projects, I have developed experience working with Python, data analysis, machine learning pipelines, and complex modeling workflows involving debugging, validation, and iterative optimization. These experiences have strengthened my ability to approach highly technical problems methodically while maintaining attention to detail and system reliability. Recently, I developed a machine learning pipeline to predict NBA game outcomes using temporal validation, feature engineering, and multiple statistical models including Logistic Regression, Random Forest, and XGBoost. While the project focused on predictive analytics, it required many of the same core skills involved in engineering verification work: structured testing, debugging, performance evaluation, validation methodology, and interpreting complex system behavior. I also completed a physics modeling project involving aircraft motion, Bayesian search theory, and ocean drift simulation, which further strengthened my experience with mathematical modeling and computational reasoning. Beyond technical work, I have held leadership positions including Student Senator, Soccer Club President, and Founder of an educational nonprofit initiative supporting STEM students from rural communities. These experiences strengthened my communication, collaboration, and organizational skills while teaching me how to work effectively within multidisciplinary teams. What excites me most about Micron's Pathfinding Design Team is the opportunity to work on early-stage memory technologies and contribute to innovative engineering solutions with real-world impact. I am especially interested in environments that combine analytical rigor, advanced technology development, and collaborative problem solving. Although my experience with semiconductor-specific verification tools is still developing, I am highly motivated to learn quickly and apply my strong quantitative background to the challenges of memory design and verification. Thank you for your time and consideration. I would welcome the opportunity to further discuss how my technical background, analytical mindset, and enthusiasm for engineering can contribute to Micron Technology. Sincerely,
Aayush Raj Pandey